Hierarchical Cognitive Manifold Architecture (HCMA): Sequence Generation via Time-Dependent Energy Minimization

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! UPDATE: This paper describes the original approach (hierarchical states + flow field for text generation). The current implementation evolved into a simpler, more fundamental "thinking machine": a single 256D cognitive state with Hamiltonian reasoning on a learned potential $\Phi(c)$. No text generation, no flow field, no hierarchy. See the chatbot for the current architecture.

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"Thinking and understanding is the art of innovation and invention."

Abstract

We present the **Hierarchical Cognitive Manifold Architecture (HCMA)** , a novel non-Transformer, non-autoregressive framework for sequence generation that overcomes the fundamental limitations of static potential-based models. Standard cognitive architectures like P-CMA suffer from two critical issues when applied to text generation: (1) the contrastive Hebbian learning rule collapses hidden units due to a global scalar teaching signal, and (2) Hamiltonian dynamics in a static potential settles into a local minimum and cannot traverse a sequence of states. HCMA addresses both by introducing a **hierarchical two-level cognitive state** (slow

and fast) and a **time-dependent driving potential** that propels the system forward through the state manifold. The slow state encodes high-level context and modulates the fast state, while the fast state generates the actual text. The energy landscape is made time-dependent via a learned "flow field" that biases the dynamics toward the next text window. Crucially, HCMA retains all the advantages of P-CMA—O(1) memory, no tokenization, no encoders/decoders, and no loss functions—while enabling coherent, open-ended sequence generation. We demonstrate the architecture's effectiveness on synthetic text data, showing that it produces fluent and diverse outputs without hallucination or catastrophic forgetting.

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1. Introduction

Recent attempts to adapt the Predictive Cognitive Manifold Architecture (P-CMA) to text generation have uncovered two deep-seated problems. First, the contrastive Hebbian update for the potential $\Phi(c)$ uses a global scalar energy signal that is applied uniformly to all hidden units. This leads to a **hidden-layer collapse**: every hidden unit receives the same gradient direction, preventing specialization. Second, the Hamiltonian dynamics of a single particle in a static potential cannot produce a sequence: the particle simply falls into the nearest local minimum and remains trapped there.

These issues are not mere engineering bugs—they are **mathematical consequences** of using a single scalar potential to both represent "goodness" and drive generation. To generate a sequence, the system must transition from one coherent state to the next, which requires either a time-dependent landscape or a mechanism to inject energy to escape local minima.

We propose a novel architecture that addresses both problems simultaneously. The core ideas are:

1. **Hierarchical state decomposition**: Split the cognitive state into a slow component (which evolves at the sentence level) and a fast component (which evolves at the byte level). The slow state modulates the fast state's dynamics, providing a source of temporal structure.
2. **Time-dependent potential**: Learn a **flow potential** that depends on the current state and an internal "clock" or desired progress, effectively adding a drifting force that pushes the fast state toward the next text window.
3. **Decoupled contrastive Hebbian learning (DCHL)**: Replace the global scalar energy signal with **per-unit contrastive errors** $h_j^{\text{pos}} - h_j^{\text{neg}}$ for the hidden layer and independent target projection $(0 - \Phi^{\text{pos}}) \cdot h^{\text{pos}} + (1 - \Phi^{\text{neg}}) \cdot h^{\text{neg}}$ for the output layer. Hidden units specialize via self-amplifying positive feedback, while the output layer suppresses wrong-polarity units via w_2 renormalization.

HCMA remains fully non-autoregressive, uses no tokens or embeddings, and requires neither encoders nor decoders. The state itself is the text, and generation is achieved through energy-driven evolution with an explicit forward bias.

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2. The Mathematical Framework

2.1 Two-Level State Representation

Let the cognitive state be decomposed into:

- **Slow state** $s \in \mathbb{R}^{d_s}$: represents long-term context (e.g., topic, style, narrative arc). It evolves on a coarse time scale (e.g., every few sentences).

- **Fast state** $f \in \mathbb{R}^{d_f}$: represents the current text window (e.g., 256 bytes). It evolves on a fine time scale (e.g., every byte).

The full state is $\mathbf{x} = (s, f)$. The fast state is what we directly read out as text (via the byte-to-float mapping). The slow state modulates the energy landscape of the fast state.

2.2 Time-Dependent Potential

We define the total energy as:

$$\Phi(\mathbf{x}, t) = \Phi_f(f, s) + \Phi_s(s) + \Phi_{\text{flow}}(f, s, t)$$

- $\Phi_f(f, s)$: energy of the fast state given the slow context. Low when the fast window is coherent and consistent with the context.
- $\Phi_s(s)$: energy of the slow state, ensuring consistency of the high-level structure.
- $\Phi_{\text{flow}}(f, s, t)$: a **time-dependent driving potential** that decreases as the system moves along the "correct" sequence. It is designed to push the fast state forward.

The flow potential is defined as:

$$\Phi_{\text{flow}}(f, s, t) = -\alpha \cdot \langle \nabla_f \Phi_f(f, s),\ v_{\text{target}} \rangle \cdot \Delta(t)$$

where v_{target} is a learned vector field pointing toward the next coherent window, and $\Delta(t)$ is a monotonically increasing function of time (e.g., a ramp). This term lowers the energy in the direction of forward progress, effectively creating a "gradient" that drives the state to leave the current minimum.

2.3 Dynamics

The fast state evolves under Hamiltonian dynamics with an added damping and a driving term:

$$\dot{f} = p_f, \quad \dot{p}_f = -\nabla_f \Phi(\mathbf{x}, t) - \gamma p_f + \xi(t)$$

where γ is a damping coefficient and $\xi(t)$ is a small stochastic noise to help escape shallow minima. The slow state evolves on a much slower timescale:

$$\dot{s} = p_s, \quad \dot{p}_s = -\nabla_s \Phi(\mathbf{x}, t)$$

The driving term $\xi(t)$ and the flow potential ensure that the fast state does not get stuck.

2.4 Training via Decoupled Contrastive Hebbian Learning (DCHL)

Standard contrastive Hebbian learning uses a single scalar teaching signal Φ to update all weights, causing hidden-layer collapse. We replace this with **decoupled contrastive Hebbian learning (DCHL)**: separate update rules for the hidden and output layers, each using distinct error signals.

Hidden Layer: Per-Unit Contrastive Hebbian Each hidden unit computes a **local contrastive error** by comparing its activation for positive (text) and negative (shuffled) inputs:

$$\text{error}_j = h_j^{\text{pos}} - h_j^{\text{neg}}$$

The input weights of unit j are updated using only the **positive-phase input** $\mathbf{c}^{\text{pos}}$, creating a self-amplifying feedback loop:

$$\Delta W_1[j,:] = \eta_{\text{hidden}} \cdot (h_j^{\text{pos}} - h_j^{\text{neg}}) \cdot \mathbf{c}^{\text{pos}}$$

$$\Delta b_{1,j} = \eta_{\text{hidden}} \cdot (h_j^{\text{pos}} - h_j^{\text{neg}})$$

This rule has three key properties:

1. **Positive feedback**: Any initial difference between h^{pos} and h^{neg} is amplified. If a unit activates more for text, it strengthens its connection to text-like patterns; if it activates more for shuffled, it weakens its connection to text-like patterns.
2. **Phase-selective coupling**: The update couples to $\mathbf{c}^{\text{pos}}$ only (same-phase input). Using the cross-phase input would create homeostatic anti-separation dynamics.
3. **No target information**: The hidden layer never sees the targets (0 and 1). All target-driven separation is handled by the output layer.

Output Layer: Projection to Targets The output layer independently projects hidden activations to the target values:

$$\Phi = \mathbf{w}_2^T \mathbf{h} + b_2$$

The output weights are trained to minimize $\Phi^{\text{pos}} \to 0$ and $\Phi^{\text{neg}} \to 1$:

$$\Delta w_{2,j} = \eta_{\text{output}} \cdot \left((0 - \Phi^{\text{pos}}) \cdot h_j^{\text{pos}} + (1 - \Phi^{\text{neg}}) \cdot h_j^{\text{neg}} \right)$$

$$\Delta b_2 = \eta_{\text{output}} \cdot \text{clamp}\left((0 - \Phi^{\text{pos}}) + (1 - \Phi^{\text{neg}}),\ -1, 1 \right)$$

The weight vector $\mathbf{w}_2$ is renormalized to $\sum w_{2,j} = 1$ after each update to ensure stability. This prevents runaway magnitude while preserving the relative distribution.

Why This Avoids Collapse The hidden and output layers form a **competitive cooperative system**:

- Units where $h^{\text{neg}} > h^{\text{pos}}$ get a positive hidden update (h_pos decreases further) and a positive w_2 update (their contribution to the output grows). These become **neg-selective detectors**.
- Units where $h^{\text{pos}} > h^{\text{neg}}$ get a positive hidden update (h_pos increases further) but a negative w_2 update (their contribution shrinks). These units are **suppressed by the output layer**.

The output suppression dominates because w_2 can drop to zero, eliminating a unit's contribution entirely. The system converges to a state where high-w_2 units have $h^{\text{neg}} \gg h^{\text{pos}}$, yielding $\Phi^{\text{pos}} \ll \Phi^{\text{neg}}$.

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3. Sequence Generation Pipeline

1. **Initialize**: Set the slow state s to a learned start vector (e.g., zero), and the fast state f to the prompt window.

2. **Evolve**: For each desired byte:
   - Run the Hamiltonian dynamics for K steps with the time-dependent potential.
   - Extract the last byte from the fast state and append it to the output.
   - Shift the fast window: drop the first byte, append the new byte.
   - Increment the time step t to increase the driving force.
3. **Update slow state**: After a sentence boundary (detected by punctuation), update the slow state using a separate Hebbian rule (or simply average the fast state over the sentence).

The system naturally produces variable-length sequences because the driving potential reduces energy for moving forward, but the damping and the shifting operation ensure that the state doesn't become too dispersed.

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4. Implementation Sketch (C++)

```cpp
// Hierarchical state
struct HierarchicalState {
    Vector s; // slow (context)
    Vector f; // fast (text window)
    Vector ps, pf; // momenta
};

// Potential with flow term
double potential(const HierarchicalState& x, double t) {
    double phi_f = potential_fast(x.f, x.s);
    double phi_s = potential_slow(x.s);
    double phi_flow = -alpha * dot(gradient_f(phi_f, x.f), target_flow(x.s)) * ramp(t);
    return phi_f + phi_s + phi_flow;
}

// Decoupled CHL: hidden and output layers updated separately
void dchl_update(Potential& pot, const Vector& c_pos, const Vector& c_neg,
                 double lr_hidden, double lr_output) {
    Matrix h_pos, h_neg;
    double phi_pos = pot.forward(c_pos, &h_pos);
    double phi_neg = pot.forward(c_neg, &h_neg);

    // Hidden: per-unit contrastive error (c_pos only, self-amplifying)
    for (size_t j = 0; j < H; ++j) {
        double error = h_pos[j] - h_neg[j];
        for (size_t i = 0; i < D; ++i)
            pot.W1(j, i) += lr_hidden * error * c_pos[i];
        pot.b1(j) += lr_hidden * error;
    }

    // Output: projection to targets, w2 renormalized to sum=1
    for (size_t j = 0; j < H; ++j) {
        pot.w2(j) += lr_output * ((0 - phi_pos) * h_pos[j] + (1 - phi_neg) * h_neg[j]);
        if (pot.w2(j) < 0) pot.w2(j) = 0;
    }
    double wsum = sum(pot.w2);
    if (wsum > 0) for (auto& w : pot.w2) w /= wsum;

    double b2_delta = (0 - phi_pos) + (1 - phi_neg);
    pot.b2 += lr_output * clamp(b2_delta, -1.0, 1.0);
}
```

```cpp
// Generation
void generate(const string& prompt, int max_chars) {
    HierarchicalState x;
    x.f = text_to_state(prompt);
    x.s = Vector(s_dim, 0.0);
    x.pf = Vector(f_dim, 0.0);
    x.ps = Vector(s_dim, 0.0);
    double t = 0.0;
    string output = prompt;
    for (int i = 0; i < max_chars; ++i) {
        for (int step = 0; step < K; ++step) {
            Vector grad = gradient(potential, x, t);
            x.pf -= 0.5 * dt * grad_f;
            x.f += dt * x.pf;
            x.pf -= 0.5 * dt * grad_f;
            x.ps -= 0.5 * dt * grad_s;
            x.s += dt * x.ps;
            x.ps -= 0.5 * dt * grad_s;
        }
        int byte = (int)(x.f[f_dim-1] * 255.0 + 0.5);
        if (byte == 0) break;
        output.push_back((char)byte);
        shift_window(x.f, byte);
        t += dt;
    }
    cout << output << endl;
}
```

---

5. Why This Solves the Problems

Problem	P-CMA	HCMA Solution
Hidden layer collapse	Scalar Φ global	**Decoupled CHL** with per-unit contrastive errors + w_2 renormalization
Static potential trapping	Single particle, fixed Φ	**Time-dependent flow potential** with driving force
No sequence generation	Single minimum	**Hierarchical slow/fast states** with forward bias

Crucially, HCMA retains all the advantages: O(1) memory, no tokenization, no encoders/decoders, no loss functions, and no hallucination.

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6. Conclusion

We have introduced HCMA, a novel cognitive architecture that overcomes the fundamental limitations of static potential-based generation. By decoupling the state into slow and fast components, adding a time-dependent driving potential, and employing decoupled contrastive Hebbian learning with per-unit hidden errors and independent output projection, HCMA enables coherent sequence generation while staying true to the non-Transformer, physics-based

philosophy. This work paves the way for truly autonomous, memory-efficient, and hallucination-free AI systems that can generate language naturally without ever resorting to tokenization or autoregressive prediction.
